

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/Georgia/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/Henry/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Static · 36 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/Josh/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/Nat/Static · 37 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/Mansh/Static · 36 trials
Total training paths	9 · 314 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	Flex
Sensor columns	20 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats → feature dim 160
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=502 · test=63

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Random Forest	0.6153	0.0656	0.9206
Stacking (SVM+RF+KNN+LogReg)	0.7149	0.0751	0.9048
LDA	0.9522	0.0183	0.7937
Voting Soft (SVM+RF+KNN+LogReg)	0.6892	0.0717	0.6984
SVM (RBF)	0.6753	0.0642	0.5079
SVM (Linear)	0.5898	0.0546	0.4762
KNN	0.3165	0.0949	0.3968
Logistic Regression	0.5799	0.0762	0.3810

Per-class hold-out F1

Class	Random Forest	Stacking	LDA	Voting (soft)	SVM (RBF)	SVM (Linear)	KNN	Logistic Regression	Support
Double_Closed_Fist	1.000	1.000	0.800	0.737	0.588	0.556	0.353	0.286	10
Double_Nothing	0.769	0.857	0.714	0.800	0.533	0.571	0.769	0.500	8
Double_Okay	0.857	0.818	0.696	0.700	0.480	0.385	0.636	0.333	9
Double_Open_Palm	0.778	0.875	0.625	0.667	0.556	0.471	0.435	0.421	9
Double_Phone	1.000	0.875	0.947	0.706	0.588	0.533	0.095	0.588	9
Double_Pistol	1.000	1.000	0.875	0.706	0.471	0.588	0.250	0.222	9
Double_Spiderman	1.000	0.900	0.889	0.588	0.353	0.316	0.286	0.333	9
macro avg	0.915	0.904	0.792	0.701	0.510	0.489	0.403	0.383	63

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.7714

Stacking (SVM+RF+KNN+LogReg)	0.7143
Voting Soft (SVM+RF+KNN+LogReg)	0.5714
SVM (RBF)	0.5429
LDA	0.5429
SVM (Linear)	0.4857
Logistic Regression	0.3714
KNN	0.1143

Per-class external accuracy

Class	Random Forest	Stacking	Voting (soft)	SVM (RBF)	LDA	SVM (Linear)	Logistic Regression	KNN	Support
Double_Closed_Fist	1.000	1.000	0.200	0.200	0.600	0.400	0.000	0.200	5
Double_Nothing	0.800	0.600	0.800	0.800	0.200	0.800	0.600	0.000	5
Double_Okay	1.000	1.000	0.800	0.800	0.000	0.800	0.600	0.000	5
Double_Open_Palm	0.000	0.000	0.200	0.200	0.800	0.200	0.200	0.000	5
Double_Phone	1.000	0.800	0.400	0.200	0.600	0.200	0.200	0.000	5
Double_Pistol	1.000	1.000	1.000	1.000	1.000	0.600	0.400	0.400	5
Double_Spiderman	0.600	0.600	0.600	0.600	0.600	0.400	0.600	0.200	5